



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	Orion: AI-Assisted Microservices-Based Online Recruitment Portal	
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Abstract:

Recruitment is an essential process for connecting students and fresh graduates with suitable employment and internship opportunities. However, the increasing number of job listings and applications makes it difficult for candidates to identify opportunities that match their actual skills and experience. Traditional recruitment platforms primarily depend on keyword-based search, which may fail to identify relevant opportunities when candidates and recruiters use different terminology to describe similar skills. Recruiters also face significant challenges in manually reviewing large numbers of applications, while candidates receive limited information about why they are considered suitable or which skills they are lacking.

To address this need, **Orion: AI-Assisted Microservices-Based Online Recruitment Portal** is proposed as a web-based recruitment platform that uses artificial intelligence to improve the recruitment process. The system enables recruiters to publish jobs and internships and allows candidates to discover and apply for suitable opportunities based on their profiles. AI is incorporated at multiple stages of the recruitment process to perform relevance-based candidate-opportunity

matching, generate explanations and skill-gap reports, and automatically rank applicants to assist recruiters in screening.

The proposed system follows a micro-services **architecture**, where independently deploy-able services communicate through REST interfaces using an API gateway. The architecture consists of services for authentication, candidate profiles and resumes, job and internship management, applications, AI-based matching and recommendations, skill-gap analysis, recruiter screening, and notifications. Each service maintains independent data and communicates through published interfaces, allowing individual components to be developed, scaled, and replaced independently.

Expected Outcome:

Orion is intended to provide a complete end-to-end recruitment workflow covering opportunity publication, opportunity discovery, AI-assisted matching, skill-gap identification, application submission and tracking, and recruiter-side applicant screening. The system focuses on improving recruitment efficiency while making AI-assisted decisions more understandable to both candidates and recruiters.

Project Architecture/Project Overview:

